

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – Intelligent Shopping Agent

Domain of Project – Business

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Problem Statement -

Problem Statement : Intelligent Shopping Agent

The Challenge

Shoppers today face an overload of product options, scattered reviews, and dynamic pricing across e-commerce platforms. Finding the right product that balances price, quality, and personal preferences is time-consuming and frustrating. Traditional recommendation systems often fail to capture real-time trends, personalized needs, or contextual factors like urgency, budget, or sustainability. The challenge is not just to recommend products but to act as a smart shopping companion that helps buyers make informed and confident decisions.

The Objective

Build an agentic AI shopping assistant that provides intelligent, personalized, and real-time support. The solution should:

Fuse Multisource Data: Aggregate product details, reviews, ratings, and price comparisons across multiple e-commerce sites into unified insights.

Enable Multimodal Search: Allow users to input voice queries, images (e.g., a photo of a product), or text descriptions to find best-matching products.

Create a Predictive & Agentic Layer: Suggest purchase timing (e.g., discounts, seasonal offers), recommend alternatives, and alert users about price drops or fake reviews.

Visualize Smart Shopping Pulse: Provide a dashboard summarizing comparisons, sustainability scores, trending products, and personalized shopping lists.

Proposed Solution -

The Intelligent Shopping Agent acts as a smart shopping companion that helps users find the best products based on their budget, preferences, and needs. It aggregates product information, reviews, ratings, and prices from multiple sources to provide unified insights. The system can predict the best time to buy, suggest alternatives, detect potentially fake reviews, and create personalized shopping recommendations. Its goal is to simplify decision-making and help users shop confidently and efficiently.

Agent 1: Smart Product Research Agent

The Smart Product Research Agent is responsible for gathering and analyzing product-related information from various sources. It compares products based on features, prices, ratings, and customer reviews. The agent identifies market trends, detects suspicious or fake reviews, and summarizes key product insights. It serves as the research and intelligence layer of the shopping system.

Agent 2: Personal Shopping Advisor Agent

The Personal Shopping Advisor Agent uses insights from the research agent to provide personalized recommendations. It considers factors such as budget, urgency, preferences, and sustainability requirements. The agent predicts optimal purchase timing, suggests alternative products, and generates a final shopping decision report. It acts as a trusted advisor that guides users toward the best purchasing choice.

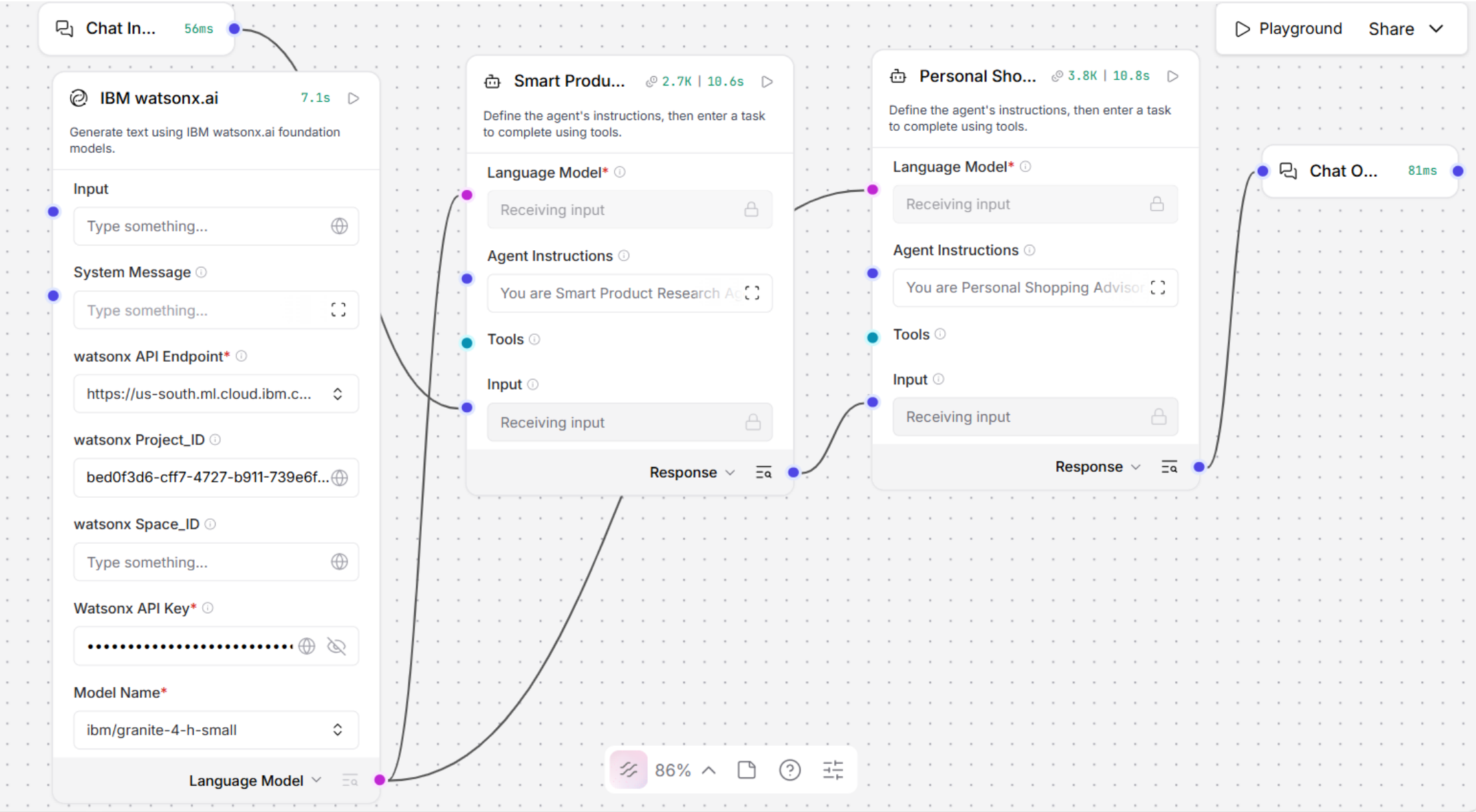
Technology Used

- **Langflow platform** -For designing and orchestrating multi-agent workflows visually.
- **IBM Granite model** - for eg – ibm-granite-4-h-small – For natural language understanding, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Intelligent Shopping Agent efficiently.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance.

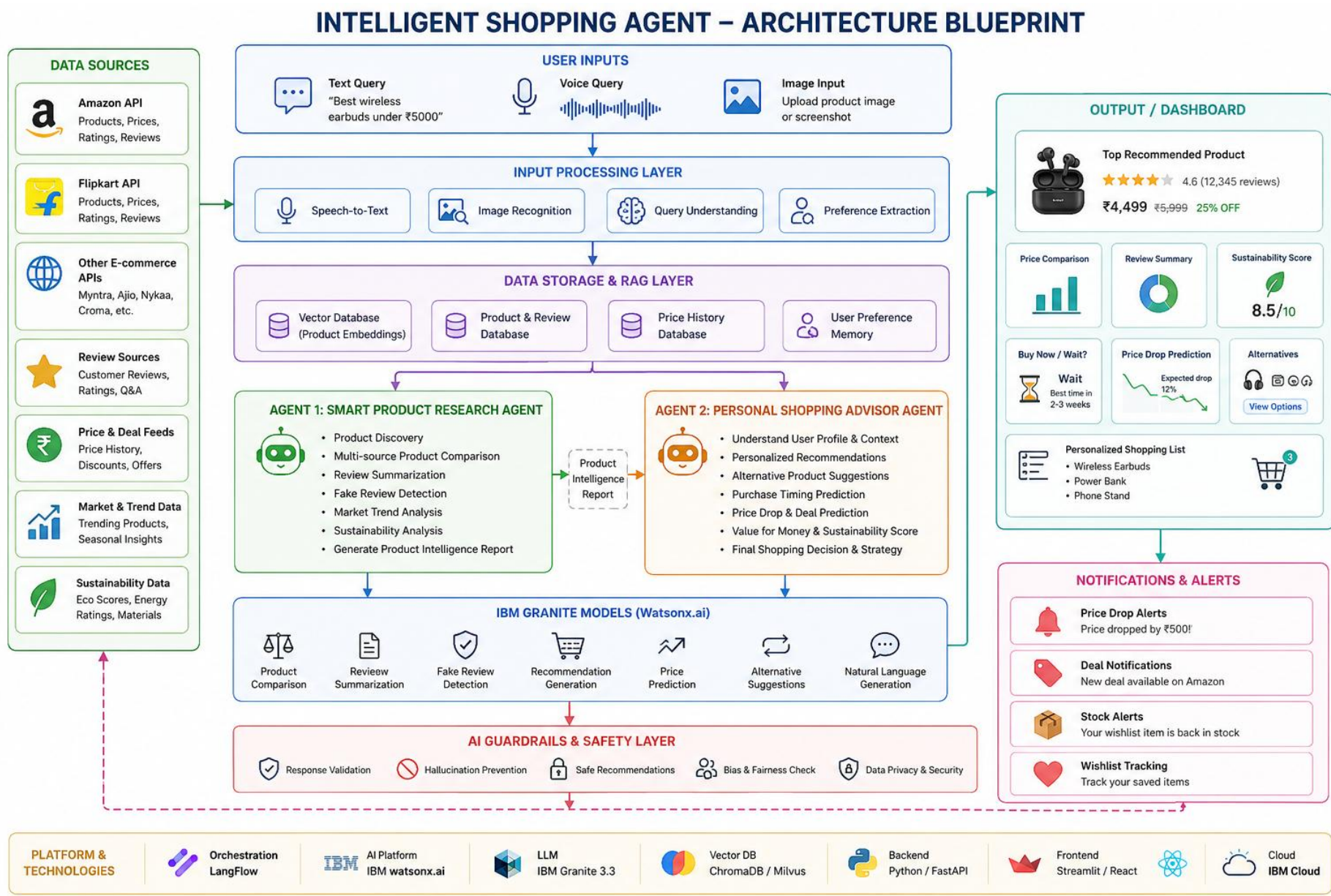
Langflow component Used -

- 1) **Chat Input** – Interface where users enter their queries, preferences, or meal details (text/voice/image).
- 2) **IBM watsonx AI Agent** - Processes user input using Granite models to generate personalized shopping insights and recommendations.
- 3) **Name of IBM Granite model** - ibm-granite-4-h-small → Fast, efficient for real-time shopping recommendations.
- 4) **Chat output** -Displays AI-generated responses such as which product to buy, comparison of products.

Langflow workflow -



Architecture blueprint

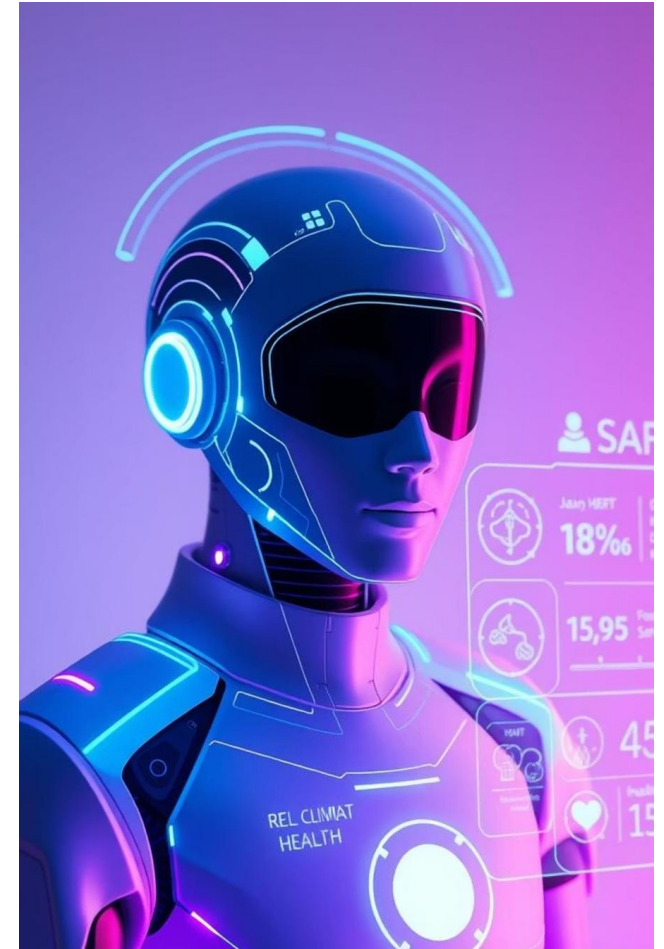


Role of Agentic AI in the solution

Role of Agentic AI in Intelligent Shopping Agent

Agentic AI enables the Intelligent Shopping Agent to act as a smart shopping companion rather than a simple recommendation system. Using IBM watsonx.ai and Granite models, multiple specialized agents collaborate to analyse products, reviews, prices, and market trends. The system understands user preferences, budget, and urgency to provide personalized recommendations, suggest alternatives, predict the best time to buy, and support informed purchasing decisions.

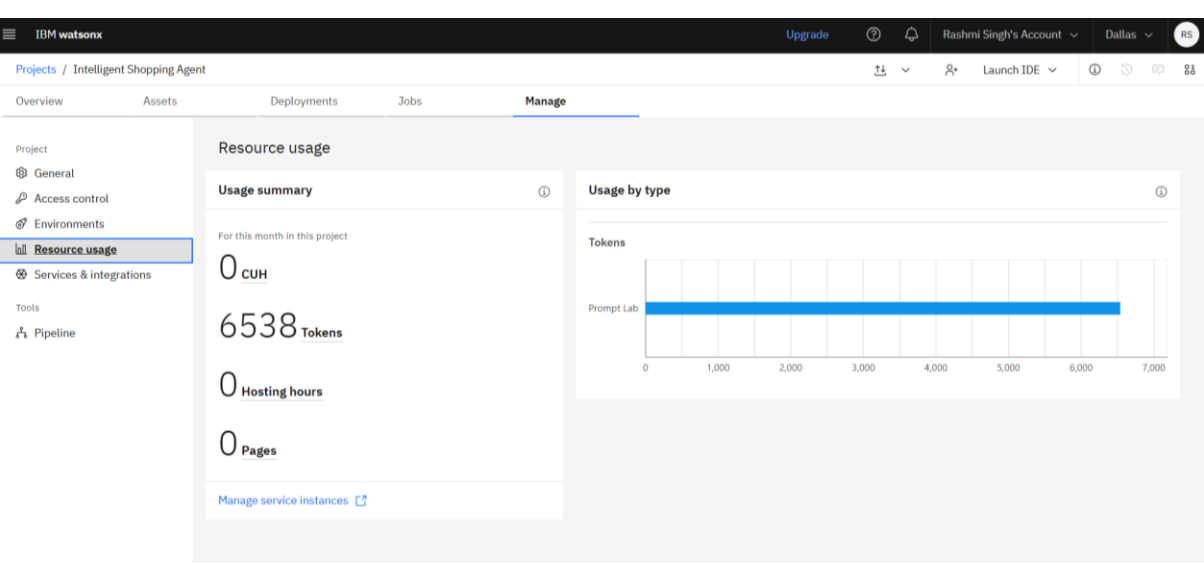
It continuously adapts to changing product information and market conditions, ensuring users receive up-to-date insights. By combining research, analysis, and decision-making capabilities, the agent helps users save time, reduce uncertainty, and make confident purchases.



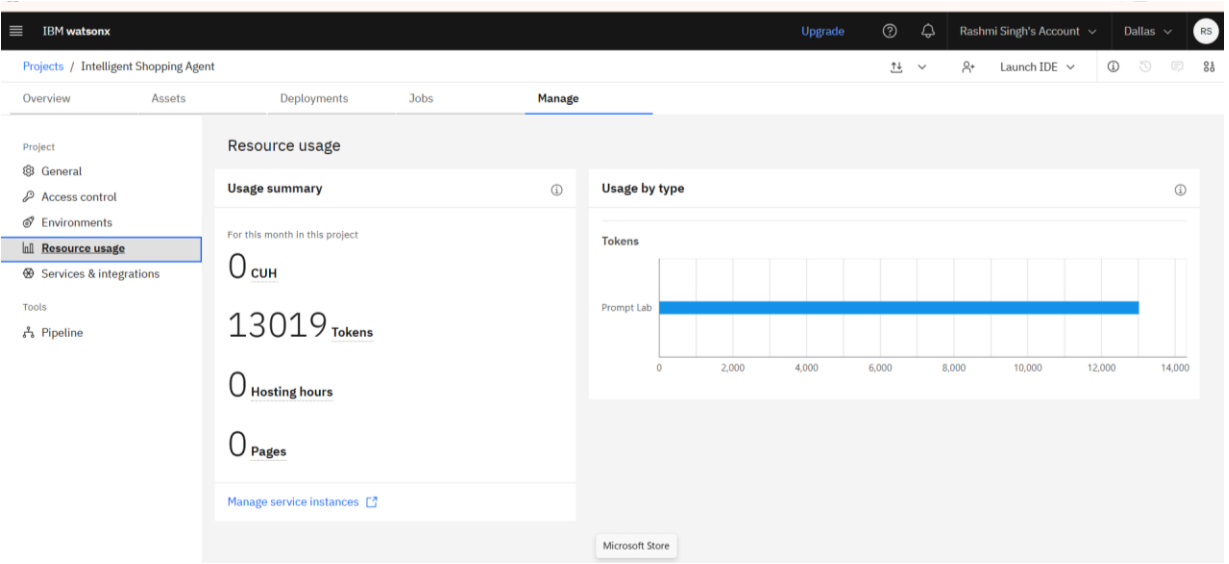
Token Usage

11,761 token are used

Before Prompt



After the Prompt



Novelty and Uniqueness

Our Intelligent Shopping Agent goes beyond traditional product recommendation systems by combining **multi-agent AI, real-time product intelligence, and personalized decision-making** in a single platform.

Unlike conventional shopping assistants, it not only compares products and analyses reviews but also detects potentially fake reviews, predicts optimal purchase timing, and recommends suitable alternatives. The system leverages **IBM watsonx.ai and Granite models** to coordinate specialized agents that collaborate like a human shopping expert and market analyst.

It supports **multimodal interactions** through text, voice, and image inputs, making product discovery more intuitive. Additionally, its sustainability scoring and price-drop prediction capabilities help users make smarter, more cost-effective, and environmentally conscious purchasing decisions.

Git Hub Link

Public github repository link –

<https://github.com/vp87rashmi-arch/FDP-IBM-Project>

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Future Scope

1. Advanced Price Prediction

The system can use historical prices and market trends to predict the best time to buy products. It can notify users about upcoming discounts and price drops. This helps users save money and make smarter purchasing decisions.

2. Hyper-Personalized Shopping Experience

The agent can learn from user preferences, shopping habits, and past purchases to provide more accurate recommendations. It can proactively suggest products that match individual needs and interests. This creates a highly personalized shopping experience.

3. Autonomous Shopping Assistant

Future versions can automatically track wishlists, monitor stock availability, and alert users about deals. Integration with e-commerce platforms and digital payments can streamline the buying process. This will make shopping faster, easier, and more convenient.

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Certificates Earned: Credly certificate



Certificates Earned: IBM BOB Certificate

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

Rashmi Chouhan

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 23 Jun 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.